Sample Question Based on Lecture 6 and Lecture 7:

Q: Which of the following statements BEST describes link encryption in communication security?

A. Encryption is performed end-to-end, ensuring data remains encrypted across all intermediate nodes.
B. Encryption is performed at the Data Link Layer, with each network link independently encrypted and decrypted at every intermediate device.
C. Only packet payloads are encrypted, while headers remain unprotected to ensure routing.
D. Encryption is handled entirely by the operating systems and applications on end hosts, requiring no changes to network devices.

Q: A financial institution deploys link encryption between its branch offices and headquarters. During an audit, it discovers that one of the routers was misconfigured and encrypts traffic only on outbound links, not inbound ones. This means frames arriving from branch locations are processed in plaintext before being passed to the next encrypted link.

What is the MOST likely security risk caused by this misconfiguration?

A. Traffic remains fully secure because encryption occurs eventually on the outbound link.
B. Traffic from branches arriving at the router is exposed in plaintext and may be read or modified before encryption occurs.
C. Only metadata is exposed, while payloads remain encrypted.
D. This configuration creates traffic loops but does not impact confidentiality.

Question: An attacker compromises a user workstation in a flat network architecture. With no segmentation, the attacker quickly gains access to file servers, development servers, and the domain controller. What architectural flaw MOST contributed to the rapid lateral movement?

A. Weak endpoint antivirus
B. Poor password policies
C. Absence of network segmentation and internal firewalls

D. No SSL inspection on perimeter devices

Q: Which of the following BEST describes the role of a network security architecture?

A. A static blueprint defining network topology and addressing plans.
B. A strategic design that integrates security controls, segmentation, and policies to protect network assets.
C. A hardware-based configuration used only for perimeter defenses.
D. An automation framework that replaces human decision-making in threat analysis.

Question: A startup grows rapidly, adding new cloud services, devices, and remote users. However, there is no structured process to update security controls or adapt to new technologies. As a result, outdated configurations persist.

Which area of security administration is MOST lacking?

A. Security Policy Definition
B. Physical Security
C. Procedures for system evolution to support continuous improvement
D. Risk Analysis

Question: A company deploys a new intrusion detection system, but after several months, leadership is unclear whether the system has improved security or not. No benchmarks or KPIs were defined before deployment.

Which principle of security planning was NOT properly applied?

A. Establish procedures for system evolution
B. Define measurable objectives to assess effectiveness
C. Implement automated alerts
D. Train the security staff

Questions: company conducts risk analysis and identifies all its critical assets. However, it proceeds directly to selecting security controls without examining how each asset might be compromised or attacked.

Question

Which fundamental step has been overlooked?

A. Determining possible vulnerabilities associated with assets
B. Surveying the costs of controls
C. Establishing policy enforcement guidelines
D. Identifying assets